

Lecture 05: Beyond the Documentation

August 26, 2024

Overview

- ▶ Interoperability
- ▶ Walkthrough

Interoperability

- ▶ Programs no longer live on their own – they often depend on many others, most notably, the operating system.
- ▶ Walkthrough deals with a relatively extreme case – no documentation at all, aside from the function name given by the export table.
- ▶ Usually some documentation is available, though it may not be sufficient
- ▶ Missing details of a contract, e.g., “Who frees the returned object?”

Undocumented APIs in Windows

- ▶ Most functions meant for access from a common, backwards-compatible, API
- ▶ Some may claim Microsoft keeps them internal to gain a competitive edge

Nevertheless, reverse engineering should be considered a last resort. In most cases getting documentation through conventional channels is preferred. On the other hand, sometimes RE can be faster.

Ground rules

In book

- ▶ Offline analysis, since examples are written, not interactive.
- ▶ No comments in assembly. This is a book on reverse engineering after all.
- ▶ Most listings are made with OllyDbg.

For class

- ▶ Mixture of dynamic and static.
- ▶ Listings all in Ghidra
- ▶ Versions will be whatever was latest at time of presentation, unless otherwise noted.

Locating Undocumented APIs

- ▶ Start with requirements. What functionality is desired or missing?
- ▶ Determine which components most likely offer or implement that functionality
 - ▶ Requires some familiarity with modules and their interfaces
 - ▶ May dump exports and search for certain keywords, e.g., “memory”, “alloc”

Walk through

Ghidra

Latest Windows or Ubuntu ("normal"), fully updated

Walk through

Please read through the text as it guides you through the reverse engineering of an undocumented API in Windows Server 2003. As you read, I recommend you follow along using a more recent version of Windows. You may use whatever tool(s) you wish for following the chapter. If you are having trouble following the control logic, try drawing control diagrams like I showed in class.

Alternatively, you can go straight to the lab materials, but there, you won't have the benefit of the author's analysis.